

Underground Relief: The Air Pollution Effects of Quito’s First Metro Line*

Leonel Borja Plaza

Inter-American Development Bank

leonelb@iadb.org

Luis Quintero

Inter-American Development Bank

luisqui@iadb.org

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Abstract

This paper studies whether the early air-quality effects of urban rail can be spatially concentrated within a city. We examine Quito Metro Line 1, Ecuador’s first metro line, which opened in December 2023 and provides direct access to the city’s historic center, a dense destination where car access is difficult. The main analysis uses weekly peak-hour data from Quito’s monitoring network and estimates station-level effects with augmented synthetic control and conformal inference. A complementary citywide analysis uses satellite data for Latin American urban areas and synthetic difference-in-differences with placebo inference. In the preferred local specification, $PM_{2.5}$ at the historic-center station, Centro, falls by 0.130 log points, equivalent to 12.2 percent, in the first months of operation, before the late-2024 energy crisis. The estimate is nearly unchanged when later post-opening weeks are added but blackout and wildfire weeks are excluded. We show that Centro is the only station with a statistically significant negative $PM_{2.5}$ estimate in a spatial placebo exercise. The response is particulate-specific: local gas pollutants do not show comparable reductions. The citywide analysis finds that satellite aerosol optical depth (AOD) moves in the same particulate direction, while satellite gas outcomes do not show a consistent pattern. The findings show that, in this setting, the air-quality benefits of a metro are concentrated at a destination where car access is difficult, a pattern that simple corridor proximity cannot explain. The results suggest that large transit investments may deliver stronger environmental gains when complemented by policies that make car access less attractive in dense destinations.

*The views expressed are those of the authors and do not necessarily reflect those of the Inter-American Development Bank, its Board of Directors, or the countries they represent.

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1 Introduction

Cities invest heavily in urban rail partly because cleaner mass transit is expected to reduce congestion, fuel combustion, and exposure to traffic-related pollution. Yet the environmental returns to these investments are not automatic. A new metro can shift some travelers away from cars and road-based public transport, but it can also free road space for other vehicles, affect some neighborhoods more than others, and operate in cities where weather, topography, and non-transport shocks also shape air quality. The relevant policy question is therefore not only whether urban rail improves air quality on average, but where within the city these gains appear and what urban conditions make them more likely.

This paper studies that question in Quito, Ecuador, using the opening of the city’s first metro line in December 2023. The setting is useful because Quito Metro Line 1 connects the city’s north-south corridor to the historic center, a dense UNESCO World Heritage site where narrow streets, limited parking, and high pedestrian activity make car access difficult. This geography creates a sharp local test. If the metro reduces pollution mainly through proximity to the rail corridor, then nearby monitoring stations should show similar changes. If instead the effect depends on destination access, then the response should be strongest where the metro changes the cost of reaching a dense, car-constrained destination. In Quito, Centro and Belisario provide this contrast. Centro, the monitoring station inside the historic center, lies 0.61 kilometers from the nearest metro stop. Belisario, outside the historic center, lies 0.77 kilometers from the nearest stop. The two stations are similarly close to the metro, but only Centro captures access to the destination where the mechanism predicts the clearest early effect.

Existing evidence shows that urban rail and related transport policies can improve air quality. [Gendron-Carrier et al. \(2022\)](#) estimate pollution effects around metro stations across cities and find larger gains in more polluted settings. Other studies examine transit expansions, traffic restrictions, or transport and emissions shocks at corridor or city scales ([Chen and Whalley 2012](#); [Li et al. 2019](#); [Goel and Gupta 2017](#); [Lin et al. 2021](#); [Chen et al. 2013](#)). We build on this literature by asking a more spatially specific question: when a large transit investment changes access to particular destinations, do pollution gains concentrate in places where car access is difficult and rail becomes a more attractive substitute? This distinction matters for both interpretation and policy. A null or modest citywide average can still mask meaningful local gains, while a local gain near a destination need not imply a smooth distance gradient around the entire corridor.

We combine two empirical designs that match these two spatial scales. The main analysis uses

weekly peak-hour readings from Quito’s ground-based air-quality monitoring network and constructs synthetic counterfactuals for exposed stations. Because the local donor pool contains only a small number of monitoring stations, the preferred local estimator is augmented synthetic control with station fixed effects, and inference relies on two-sided conformal p -values. We then place the local evidence in broader context using satellite retrievals for Latin American urban areas. The citywide analysis compares Quito to similar cities using synthetic difference-in-differences. The two designs therefore answer different questions: the local design tests whether pollution changed at the predicted destination, while the citywide design asks whether Quito experienced a detectable aggregate change relative to a counterfactual constructed from comparable urban areas.

The results show a localized particulate response at Quito’s historic center. In the preferred local specification, $PM_{2.5}$ at Centro falls significantly, by about 12 percent, during the first months of operation, before Ecuador’s late-2024 blackout and wildfire disruptions. The estimate is nearly unchanged when later post-opening weeks are added, as long as the blackout and wildfire weeks are excluded. Belisario, the similarly close station outside the historic center, does not show comparable evidence of a post-opening $PM_{2.5}$ reduction. A leave-one-station-out spatial placebo exercise shows that Centro is the only station with a statistically significant negative $PM_{2.5}$ estimate during this early period. The local pollutant pattern is also particulate-specific: NO_2 is a well-fit null, CO is uninformative, and SO_2 turns positive only when weeks affected by the energy crisis enter the estimation window.

The citywide satellite evidence points in the same particulate direction, but we interpret it as complementary rather than as the paper’s main causal test. In the preferred 100-city donor pool, aerosol optical depth falls by 0.113 log points during the first months of operation, before Ecuador’s late-2024 blackout and wildfire disruptions. The estimate is also negative across alternative donor pools. Satellite gas outcomes do not show a consistent pattern. These results suggest that the local particulate finding is not contradicted at the city scale, but they do not establish the mechanism. Aerosol optical depth measures aerosol loading in the atmospheric column rather than surface $PM_{2.5}$, and the citywide comparison is more exposed to national shocks, regional atmospheric conditions, and non-transport sources.

The paper makes two contributions. First, it shows that the early air-quality benefits of a metro can be concentrated at a destination where car access is difficult, rather than following simple proximity to the rail corridor. This complements evidence on average pollution effects around transit stations and at the city scale by showing why the location of effects within the city matters. The result also has a direct policy implication: large transit investments may generate stronger environmental gains when they are complemented by measures that make car use less attractive in dense destinations. Second, the paper illustrates an empirical strategy for studying transport investments when exposure is local but citywide shocks are important. A small-donor local synthetic-control design with conformal

inference provides the main evidence, while the citywide comparison asks whether Quito’s aggregate pollution changed relative to similar Latin American cities after the metro opened. The findings document where the particulate response appears and are consistent with destination-based mode substitution, while leaving the direct traffic channel for future evidence on ridership, congestion, and vehicle flows.

The rest of the paper proceeds as follows. Section 2 describes the institutional setting. Section 3 describes the data and estimation samples. Section 4 presents the empirical strategy. Section 5 reports the local and citywide results. The final section discusses interpretation and implications for transport policy and air-quality evaluation.

2 Institutional Background

Quito is located in a narrow Andean valley on the eastern slopes of the Pichincha volcano, at roughly 2,850 meters above sea level. Its topography matters for air quality. Steep terrain, frequent temperature inversions, and stagnant-air episodes can limit dispersion and trap pollutants near the surface ([Encyclopaedia Britannica 2026](#); [Cazorla 2016](#); [Zalakeviciute et al. 2024](#)). These conditions make traffic-related emissions especially relevant for local exposure, particularly for fine particulate matter. Prior air-quality research on Quito emphasizes that basin topography, diesel traffic, and meteorological conditions have contributed to persistent particulate pollution above international health guidelines ([Cazorla 2016](#); [Zalakeviciute et al. 2024](#)).

Quito’s historic center is central to the paper’s mechanism. The Centro Histórico is a UNESCO World Heritage site, with dense activity, narrow colonial streets, limited parking, and high pedestrian flows ([Encyclopaedia Britannica 2026](#); [UNESCO World Heritage Centre n.d.](#)). These features make automobile access costly even though the area is geographically central. A metro can therefore affect pollution in the historic center by moving travelers along a corridor, and also by changing the relative cost of reaching a dense destination where car access is difficult. This distinction motivates the empirical design: Centro and Belisario are both close to Metro Line 1, but only Centro is located in the historic center.

Quito Metro Line 1 is the city’s first metro line. Commercial service began on December 1, 2023, after earlier trial operations were suspended. The line runs entirely underground for about 22.6 kilometers along Quito’s north-south valley corridor, connecting Quitumbe in the south with El Labrador in the north through 15 stations ([Transdev 2023](#)). The end-to-end trip takes roughly 34 minutes, and the system was planned for several hundred thousand daily passengers once mature ([Oliver Wyman Forum 2024](#)). One station, San Francisco, is located in the historic center, providing direct metro access to the area monitored by the Centro station. Because the system opened only at the end of 2023 and ridership was still ramping up during 2024, the estimates in this paper capture early operational effects rather than long-run impacts.

The first year of operation also coincided with an unusually disruptive national context. In 2024, Ecuador experienced a severe drought that constrained hydroelectric generation and led to scheduled electricity rationing beginning in September. The crisis intensified in October, and households and firms responded by using diesel generators, a source of local combustion emissions unrelated to the metro. Wildfires around Quito during the same period further degraded air quality (Vallejo et al. 2025; Zalakeviciute et al. 2024). This institutional shock is important for the empirical design. It motivates estimating a pre-disruption window that ends before the blackout and wildfire period, a donut window that excludes the disruption weeks, and a full window that shows how estimates change when those weeks are retained.

3 Data

We assemble two panels at two spatial scales. The first is a panel of ground-based air-quality monitors in Quito. It measures surface pollution at fixed monitoring stations across the metropolitan area and provides the main local evidence in the paper. The second is a panel of satellite retrievals across Latin American urban areas. The city-level panel treats Quito’s transport system as a whole, recognizing that a metro can affect surface traffic near stations and also indirectly across the broader urban network. Given the broader Ecuadorian context of 2024, which combined a security emergency, fiscal contraction, and a drought-driven energy crisis that may confound city-level comparisons, we interpret the citywide analysis as complementary to the local station analysis rather than as the paper’s primary estimate of exposure changes.

We describe the treatment timing and the 2024 disruption that shapes the estimation windows (Section 3.1), the air-quality monitoring station panel (Section 3.2), and the satellite-derived urban-center panel (Section 3.3). We discuss the estimators in the empirical strategy section.

3.1 Treatment timing and the 2024 disruption

Quito Metro Line 1 opened to passengers on Friday, December 1, 2023. Treatment is defined at the weekly level, with the post-opening indicator turning on in the ISO week¹ that contains the opening date, beginning November 27, 2023.

The post-opening period is heterogeneous, so we divide it into three blocks. The first, P_1 , runs from the opening week through the week beginning September 9, 2024. The second is a disruption block spanning the weeks beginning September 16 through December 16, 2024, when national electricity rationing pushed households and firms onto diesel generators and wildfires burned around the capital. These shocks increased ambient pollution in Quito for reasons unrelated to the metro (Vallejo et al. 2025; Zalakeviciute et al. 2024). The final post-opening block, P_2 , begins

¹ISO weeks follow the international week-date convention: weeks begin on Monday, and week 1 is the week containing the first Thursday of the calendar year. Under this convention, the week containing December 1, 2023 begins on Monday, November 27, 2023 and is labeled ISO 2023-W48.

Table 1. Local and citywide analyses at a glance

	Local ground-monitor analysis	Citywide satellite analysis
Treated unit	REMMAQ monitoring stations, with Centro as the main treated station	Quito urban center, UCDB identifier 2544
Comparison pool	Other REMMAQ stations reporting the same pollutant	Latin American urban areas ranked by similarity to Quito
Geographic unit	Monitoring station	UCDB urban-center polygon
Pollutants	PM _{2.5} , CO, NO ₂ , SO ₂	AOD, CO, NO ₂ , SO ₂
Temporal resolution	Weekly averages of weekday peak-hour readings	Weekly satellite retrievals, aggregated to four-week blocks for estimation
Pre-treatment period	52 weeks before the opening week	100 weeks before the opening week
Post-opening windows	Pre-disruption, donut, and full	Pre-disruption, donut, and full
Panel size before estimation	PM _{2.5} : 8 stations $\times$ 116 weeks; gases: 7 stations $\times$ 134 weeks	Pollutant-specific balanced panels, with 227 weekly observations per city
Primary role	Main test of localized air-quality effects near the historic center	Citywide benchmark for aggregate pollution changes in Quito

Local PM_{2.5} covers all eight monitoring stations. CO, NO₂, and SO₂ cover seven stations because San Antonio does not report those series. The satellite panels are balanced separately by pollutant.

in January 2025. The two weeks beginning December 23 and December 30, 2024 fall after the main rationing period but before the beginning of P_2 , and are kept in the full post-opening window.

We use this partition to define three estimation windows. The pre-disruption window uses only P_1 . The donut window drops the disruption weeks and retains the post-opening weeks before and after the disruption. The full window retains every post-opening week. These are not data-cleaning exclusions. They are estimand choices that allow us to distinguish the early post-opening period from a later period contaminated by large citywide shocks. Table 1 summarizes the two empirical designs.

3.2 Local analysis: ground-based air monitoring data

The REMMAQ² data consist of validated hourly station readings published by Quito’s municipal environmental authority (REMMAQ 2026). They cover pollutant concentrations and meteorological conditions across the metropolitan area. We use all eight stations with available data for the analysis period: Belisario, Carapungo, Centro, Cotocollao, Guamaní, Los Chillos, San Antonio, and Tumbaco. We study four pollutants, PM_{2.5}, CO, NO₂, and SO₂, and use temperature, humidity, wind speed, wind direction, rainfall, solar radiation, and surface pressure as meteorological controls.

Two stations are especially important for the design. Centro is located inside Quito’s historic center, where narrow streets, limited parking, and high pedestrian activity make automobile access unusually costly. Belisario is located near the Metro corridor but outside the historic center. Centro is 0.61 km from the nearest Line 1 stop, while Belisario is 0.77 km away. The two stations are therefore similarly close to the metro but differ sharply in the urban setting most relevant to the mechanism.

²Quito’s Metropolitan Atmospheric Monitoring Network.

This geography helps separate a simple corridor-proximity interpretation from a destination-access interpretation centered on the historic center. Figure 1 maps the REMMAQ stations and Line 1 corridor and stations.

PM_{2.5} is the preferred local outcome because all eight monitoring stations report it, preserving the largest donor pool for the station-level counterfactual. CO, NO₂, and SO₂ provide robustness and mechanism checks, but each is estimated with a smaller donor pool because San Antonio does not report those series. Table 2 lists the stations, their distance to the nearest stop, and the pollutant series each reports.

Table 2. Air-quality monitoring stations

Station	Distance (km) ^a	PM _{2.5}	CO	NO ₂	SO ₂
<i>Centro</i>	0.61	✓	✓	✓	✓
<i>Belisario</i>	0.77	✓	✓	✓	✓
<i>Guamaní</i>	4.04	✓	✓	✓	✓
<i>Cotocollao</i>	5.21	✓	✓	✓	✓
<i>Carapungo</i>	7.64	✓	✓	✓	✓
<i>Los Chillos</i>	8.62	✓	✓	✓	✓
<i>Tumbaco</i>	9.56	✓	✓	✓	✓
<i>San Antonio</i>	16.65	✓			
Stations in balanced panel		8	7	7	7

^a Straight-line distance from the monitoring station to the nearest Metro Line 1 stop.

Source: REMMAQ, Secretaría de Ambiente, Municipio de Quito. Stations are ordered by distance to the nearest metro stop. Centro is the main treated station in the local analysis. Belisario is the nearest station and the key comparison for distinguishing proximity from destination access. San Antonio reports PM_{2.5} only.

We restrict the hourly data to weekday morning and evening peak hours, 7–9 a.m. and 5–7 p.m., when traffic-related emissions are most likely to contribute to short-run variation in local pollution. For each station and week, we average the peak-hour readings into a weekly mean concentration and use the natural log of that weekly mean as the outcome. The PM_{2.5} panel begins in the week of November 28, 2022 and runs through March 2025, yielding 116 balanced station-weeks per station. The gas panels run through June 2025, yielding 134 balanced station-weeks per station.

Because synthetic-control estimators construct donor weights by matching the treated unit's pre-treatment trajectory, the optimization requires a rectangular panel with no missing station-week observations in the estimation window. We therefore use limited imputation only to recover otherwise usable short gaps before balancing the panel. Short gaps of one or two weeks are linearly interpolated within station. Longer gaps are left unfilled, except for a longer San Antonio PM_{2.5} segment that is predicted using other stations and weather covariates.³ Imputation is limited in the balanced panels: 3.0 percent of PM_{2.5} observations, 1.0 percent of CO observations, 0.7 percent of NO₂

³The prediction regression excludes Centro and Belisario, the two stations closest to the line, so that imputed control values do not mechanically incorporate changes from the locations most likely to reflect metro-related effects.

observations, and 1.3 percent of SO₂ observations. Centro has no imputed PM_{2.5} observations. Table 3 summarizes the balanced panels and estimation windows. The full sample-construction funnel is reported in Appendix A.

Table 3. Estimation samples: balanced weekly panel composition by pollutant

Pollutant	Stations	Post-treatment weeks					Estimation samples (weeks)		
		Pre	P1	Blackout	Trans.	P2	<i>pre_blackout</i>	<i>donut</i>	<i>full</i>
PM _{2.5} ^a	8	52	42	14	2	6	94	102	116
CO	7	52	42	14	2	24	94	120	134
NO ₂	7	52	42	14	2	24	94	120	134
SO ₂	7	52	42	14	2	24	94	120	134

Notes. Balanced weekly panels of peak-hour, weekday pollutant concentrations. Peak hours are 07:00–09:00 and 17:00–19:00. The pre-treatment period contains 52 weeks for all pollutants; treatment begins in ISO 2023-W48, the week containing the December 1, 2023 opening of Quito Metro Line 1. “Blackout” denotes the Phase 3 power-outage and wildfire disruption weeks, which are excluded from the *donut* sample and occur after the *pre_blackout* sample ends. “Trans.” is the post-outage transition fortnight in late December 2024. *pre_blackout* equals pre-treatment plus P1; *donut* equals the full sample minus blackout weeks; *full* uses all weeks. San Antonio is unavailable for CO, NO₂, and SO₂ and is dropped from those panels. Short station-level gaps of up to two weeks are linearly interpolated, and San Antonio’s PM_{2.5} series is regression-imputed where needed; these observations are filled values, not absent weeks.

^a PM_{2.5} contains a 7-week interior gap, from the week of January 13 to the week of February 24, 2025, that is absent from the balanced panel for all stations, so the PM_{2.5} panel jumps from the week of January 6, 2025 to the week of March 3, 2025. This accounts for PM_{2.5}’s shorter Period 2. CO, NO₂, and SO₂ have contiguous weekly sequences.

Figure 2 plots the weekly pollutant series around the opening. The figure is descriptive, but it previews two design features that guide the empirical analysis. First, Centro and Belisario are the two closest stations to Line 1, yet they do not move together uniformly after the opening. Second, the late-2024 disruption generates visible changes in several series, motivating the use of separate post-opening windows rather than a single average over the entire post-treatment period.

3.3 City-level analysis: satellite retrievals

The city-level panel uses satellite retrievals to compare Quito with other Latin American urban areas. This analysis asks whether Quito experienced a detectable aggregate change in pollution relative to comparable cities after Metro Line 1 opened.

We define cities using the Global Human Settlement Urban Centre Database, 2024A release, which provides consistently delineated urban-center polygons across countries (Schiavina et al. 2024).⁴ The Latin American UCDB sample contains 1,036 urban areas. Quito is UCDB identifier 2544 and is the treated city. We retrieve and aggregate weekly pollutant measures for each urban center through Google Earth Engine (Gorelick et al. 2017), and merge them with weekly meteorological covariates from ERA5-Land (Muñoz-Sabater et al. 2021).⁵ The weather controls include

⁴The UCDB follows the Degree of Urbanisation methodology, which defines urban centers as high-density population clusters using harmonized grid-cell criteria rather than administrative boundaries (Dijkstra et al. 2021).

⁵For each pollutant and ISO week, we compute the mean of all valid satellite pixels intersecting each UCDB urban-center polygon. Product-specific quality filters are applied before aggregation. The resulting observation is therefore a weekly urban-area average of valid retrieved pixels.

Table 4. Citywide satellite panels

Outcome	Product	Transform	Balanced cities	Weeks per city	Quito imputed weeks	Pre-treatment diff.
AOD	MODIS MAIAC	Log	572	227	6	0.196
CO	S5P TROPOMI	Log	1,035	227	0	-0.386
NO ₂	S5P TROPOMI	Log	906	227	7	0.616
SO ₂	S5P TROPOMI	Asinh	861	227	15	-0.050

The initial satellite sample contains 1,036 Latin American urban areas and 227 weekly observations per city. Balanced cities are retained after pollutant-specific coverage, short-gap imputation, and balance requirements. Quito imputed weeks report the total number of pre- and post-opening weeks imputed for the treated city. The pre-treatment difference is the normalized difference between Quito and the preferred 100-city donor pool in the transformed outcome mean before the opening week.

temperature, relative humidity, surface pressure, precipitation, solar radiation, evaporation, soil temperature and moisture, and wind conditions.

The satellite outcomes are AOD, CO, NO₂, and SO₂. AOD comes from the MODIS MAIAC MCD19A2 product at ~ 1.1 km resolution (Lyapustin et al. 2018). It measures aerosol loading in the atmospheric column and is a proxy for particulate pollution.⁶ CO, NO₂, and SO₂ come from Sentinel-5P TROPOMI (Veefkind et al. 2012). CO and NO₂ are traffic-related atmospheric indicators, while SO₂ is used mainly as a falsification outcome because urban SO₂ is less directly tied to passenger transport.

We process all satellite products to weekly city-level means and then construct balanced pollutant-specific panels. AOD, CO, and NO₂ are transformed using logs. SO₂ is transformed using the inverse hyperbolic sine because TROPOMI SO₂ retrievals can be zero or negative over low-emission cities (Bellemare and Wichman 2020); these values are valid noisy retrievals rather than missing observations. As in the local analysis, the synthetic-control estimators require a rectangular city-week panel: each donor city must have a valid outcome in every week used for estimation. Short gaps of one or two weeks are imputed before balancing. The balanced panels retain 572 cities for AOD, 1,035 for CO, 906 for NO₂, and 861 for SO₂, each with 227 weekly observations per city. Table 4 summarizes the satellite data used in the citywide analysis.

For donor-pool construction, we rank Latin American urban areas by Mahalanobis distance to Quito using pre-treatment and time-invariant city characteristics that affect pollutant levels, atmospheric dispersion, and satellite retrieval.⁷ Restricting the donor pool to cities closest to Quito in pre-treatment characteristics reduces the scope for overfitting in synthetic-control designs (Abadie and Vives-i Bastida 2024). The estimation uses a sequence of donor pools, including the 50, 100,

⁶In Quito’s high-altitude basin, thermal inversions and vertical mixing can weaken the relationship between column aerosol loading and surface exposure (Boyouk et al. 2010; Cazorla 2016).

⁷The Mahalanobis distance is computed using the extended UCDB covariate set: elevation; annual mean temperature; temperature seasonality; annual precipitation; log population; log urban area; road density; average building height; NOx emissions per capita; PM_{2.5} emissions per capita; the transport share of NOx emissions; and average GDP.

150, and 200 nearest cities, as well as balanced panels with all available cities. The preferred citywide specification uses the 100 nearest donor cities, while the alternative donor pools are reported as robustness checks. The last column of Table 4 reports the normalized pre-treatment difference in the transformed outcome between Quito and the preferred donor pool. Full pre-treatment outcome and weather-covariate diagnostics are reported in Appendix A.1.

Figures 3 and 4 summarize the citywide panel. The outcome figure shows that Quito’s satellite series are more volatile than the donor-pool median, particularly for NO₂ and SO₂, which motivates interpreting the satellite estimates as a broad aggregate benchmark rather than as the primary test of local exposure changes. The coverage figure shows that donor-pool coverage is high for CO and generally high for NO₂ and SO₂, while AOD has lower and more variable coverage. Quito’s missing weeks are limited and are filled only through short-gap imputation. Detailed coverage, imputation, and balance diagnostics are reported in Appendix A.1.

4 Empirical Strategy

We estimate the effect of Quito’s first metro line on air pollution by constructing counterfactual pollution paths for exposed locations after the start of commercial operations. Let Y_{it} denote pollution in unit i in week t , where the unit is a REMMAQ monitoring station in the local analysis or an urban area in the citywide analysis. For each treated unit, the object of interest is the average post-opening difference between observed pollution and the pollution path that would have occurred absent the metro:

$$\tau_i = \frac{1}{T_1} \sum_{t \geq T_0} \left(Y_{it} - \hat{Y}_{it}(0) \right),$$

where τ_i is the treatment effect, T_0 is the week of the metro opening, T_1 is the number of post-opening periods used in the estimate, and $\hat{Y}_{it}(0)$ is a synthetic counterfactual constructed from units not assigned to treatment. The identifying assumption is that, after conditioning on pre-opening pollution dynamics and weather, the synthetic comparison path approximates the untreated potential outcome of the treated unit. This logic follows the synthetic-control tradition of constructing a comparison path from weighted untreated units rather than relying on a single untreated comparison unit (Abadie et al. 2010).

The local station analysis is designed to distinguish two mechanisms. A corridor-proximity mechanism predicts that pollution changes should be strongest at monitors close to Metro Line 1. The destination-access mechanism predicts that effects should be more pronounced where the metro changes the relative cost of reaching the historic center. This distinction is central in Quito because Centro and Belisario are both close to Line 1, but only Centro lies inside the historic center, where narrow colonial streets, limited parking, and high pedestrian activity make car access unusually costly. We therefore first examine whether the data support a proximity interpretation. Belisario

provides the key discriminating comparison: if proximity to the line is sufficient to reduce pollution, Belisario should also experience a decline; if the effect operates through mode substitution for trips bound for the historic center, Belisario should not.

We operationalize this test using the other REMMAQ stations as the local comparison group. For each station, we construct a synthetic counterfactual from the remaining stations and examine whether the post-opening deviation is unusual. This leave-one-out exercise serves two purposes. First, it tests whether effects trace a general proximity pattern across the monitoring network. Second, it verifies whether Belisario behaves as an untreated comparison station before it is used as part of the donor pool for Centro. Thus, the role of Belisario is not circular: its null response is first a diagnostic against the corridor-proximity hypothesis and then a validation of its inclusion as a donor for the Centro counterfactual.

We implement two related counterfactual estimators. The first is augmented synthetic control, implemented with the `augsynth` package (Ben-Michael et al. 2021a,b). The second is synthetic difference-in-differences, implemented with the `synthdid` package (Arkhangelsky et al. 2021, 2024). Both estimators construct a counterfactual pollution path using untreated comparison units, but they differ in how they use pre-treatment information, how they incorporate covariates, and in the type of inference that is most credible in our two empirical settings.⁸

Augmented synthetic control starts from the synthetic-control idea of approximating the treated unit with a weighted average of untreated units, but adds an outcome model to correct remaining pre-treatment imbalance (Ben-Michael et al. 2021a). Let X_1 denote the vector of pre-treatment outcomes and covariates for the treated unit, and let X_0 denote the corresponding matrix for donor units. Standard synthetic control chooses weights $\hat{w}$ to make $X_0\hat{w}$ close to X_1 . Augmented synthetic control adjusts the post-treatment synthetic-control contrast by an estimate of the remaining bias:

$$\hat{\tau}_t^{\text{aug}} = Y_{1t} - \sum_{j \in \mathcal{D}} \hat{w}_j Y_{jt} - \left[\hat{m}(X_1) - \sum_{j \in \mathcal{D}} \hat{w}_j \hat{m}(X_j) \right],$$

where $\mathcal{D}$ is the donor pool and $\hat{m}(\cdot)$ is an outcome model estimated from pre-treatment data. In our preferred local implementation, $\hat{m}(\cdot)$ is estimated using ridge regression with station fixed effects. The fixed effects absorb persistent level differences across monitors, while the ridge outcome model uses pre-treatment variation to adjust remaining imbalance between the treated station and

⁸The two estimators handle controls differently. The `synthdid` implementation operates on an outcome matrix and does not directly include covariates, so we partial out weather controls and calendar effects before estimation and run SDID on the adjusted outcome. In the local station analysis, this adjustment residualizes log pollution on weather controls and month indicators, estimated on the control stations, excluding both Centro and Belisario. In the citywide satellite analysis, we use an analogous split-FWL procedure: the SDID outcome is fully residualized on weather covariates and urban-area-by-month fixed effects, while the AugSynth outcome is adjusted only for those fixed effects and the residualized covariates are passed directly to the ridge outcome model. AugSynth can incorporate covariates directly, so the ridge adjustment learns the covariate-outcome relationship within the estimator.

its synthetic counterfactual. The ridge adjustment is useful because exact pre-treatment balance is difficult when the donor pool is small. It improves fit while regularization limits extrapolation and overfitting.

Synthetic difference-in-differences combines synthetic-control unit weights with difference-in-differences time weights (Arkhangelsky et al. 2021). In a single-treated-unit setting, the estimator can be written intuitively as

$$\hat{\tau}^{\text{sdid}} = \left(\bar{Y}_{1,\text{post}} - \sum_{j \in \mathcal{D}} \hat{w}_j \bar{Y}_{j,\text{post}} \right) - \sum_{t < T_0} \hat{\lambda}_t \left(Y_{1t} - \sum_{j \in \mathcal{D}} \hat{w}_j Y_{jt} \right),$$

where $\hat{w}_j$ are donor weights and $\hat{\lambda}_t$ are pre-treatment time weights. The first term compares the treated unit to a synthetic control after the metro opens. The second term subtracts the weighted pre-treatment gap between the treated unit and the same synthetic control. This makes SDID less dependent on exact pre-treatment matching than standard synthetic control, while retaining a data-driven comparison group.

The two estimators play different roles because the comparison sets differ sharply across analyses. In the local REMMAQ analysis, the donor pool contains only the other Quito monitoring stations. With such a small donor pool, the inference procedures recommended for SDID in larger panels are not reliable as the main inferential basis: placebo and large-sample approximations have very little information when there are only a few untreated units. Accordingly, local SDID results are reported only as point-estimate robustness checks, without p -values. We therefore use augmented synthetic control with station fixed effects as the preferred local estimator and use SDID as a robustness check. This local hierarchy is design-based: augmented synthetic control gives a transparent station-level counterfactual, and conformal inference can be applied directly to the prediction errors of that counterfactual.

Inference for the local augmented synthetic-control estimates is based on conformal methods for counterfactual and synthetic-control settings (Chernozhukov et al. 2021). The intuition is simple. Under the null hypothesis of no treatment effect, the post-opening prediction error should not look unusually large relative to the prediction errors observed before the metro opened. Conformal inference tests candidate treatment effects by shifting the post-treatment outcomes and asking which shifts make the full sequence of prediction errors look consistent with the pre-treatment errors. This is attractive in the local setting because it does not require a large number of donor stations.

We use a leave-one-out spatial placebo exercise to assess whether the conformal procedure over-rejects and whether the estimated reduction is specific to Centro. We shift treatment status across the monitoring network by treating each station in turn as if it were exposed, construct its synthetic counterfactual from the remaining stations, and apply the same conformal inference

procedure. This exercise is not the main source of statistical inference for Centro. With only eight stations, any rank-based comparison is coarse and has a natural lower bound of $1/8$. Instead, it checks the design’s internal logic: the method should not find significant reductions at stations where the destination-access mechanism predicts none, and the reduction should appear only at Centro. Belisario is the key placebo because it is nearly as close to Line 1 as Centro but outside the historic center.

As described in the data section, we estimate effects on three post-opening windows for all pollutants. The full post-opening window keeps all available weeks after the metro began operating, including the late-2024 disruption period. We also report a clean pre-disruption window because Ecuador’s energy crisis generated citywide pollution shocks that could mask or confound metro-related changes in air quality. This is not a sample-cleaning exclusion, but an estimand choice: it isolates the period after the metro opened and before the major citywide disruption. Alternative windows show how the estimates change when the disruption period and the more limited post-disruption coverage are included. PM_{2.5} is the preferred local outcome because all eight monitoring stations report it, preserving the largest donor pool. Other pollutants provide robustness and mechanism checks, but are estimated with a smaller donor pool because San Antonio does not report those series.

The satellite analysis estimates a separate, city-level estimand. Instead of comparing monitors within Quito, it compares Quito as an urban area to a synthetic combination of other Latin American urban areas. This design asks whether the metro generated a detectable aggregate change in pollution at the city scale. It is not designed to detect localized changes inside the historic center. In this setting, the comparison set is much larger. Quito is compared to a donor pool of Latin American urban areas ranked by similarity, and the primary specification uses synthetic difference-in-differences. This choice follows the structure of the method: with many candidate donor cities and repeated pre-treatment observations, SDID can use both unit weights and time weights to construct an aggregate counterfactual for Quito ([Arkhangelsky et al. 2021](#)).

For SDID and synthetic-control estimates in the satellite analysis, we report placebo standard errors and Wald-type tests, complemented by rank-based placebo diagnostics. The placebo standard errors are computed by repeatedly reassigning treatment to donor units and using the dispersion of those placebo estimates as the sampling variability of the estimator. The Wald tests then compare the estimated effect to this placebo-based standard error. This approximation is more informative in the city-level analysis than in the local station analysis because the satellite donor pool is large enough to generate a meaningful distribution of placebo estimates. We retain rank-based permutation p-values as a secondary check reported in the appendix; they lead to the same substantive conclusions.

We also estimate augmented synthetic-control specifications in the satellite analysis, but interpret them as robustness checks. The same flexibility that makes augmented synthetic control useful

in the local station setting can become less desirable when the donor pool is large and the pre-treatment satellite panel is high-dimensional. In that setting, ridge regularization helps, but we still prefer SDID as the primary estimator because it imposes a more transparent weighting structure for the aggregate city-level comparison. The hierarchy is therefore reversed across the two analyses: augmented synthetic control is primary locally and robustness at the city level, while synthetic difference-in-differences is primary at the city level and robustness locally.

The two designs answer different questions. The satellite analysis asks whether Quito experienced a detectable citywide pollution change relative to comparable Latin American urban areas. The local station analysis asks whether $PM_{2.5}$ changed at the car-constrained destination where the mechanism predicts the strongest early effect, and whether the same pattern is absent at an almost equidistant non-historic center monitor. The empirical strategy therefore does not require a smooth distance gradient from the metro corridor. With only eight monitoring stations, a short post-opening period, and a major late-2024 disruption, the local design is better suited to testing the sharper prediction that the response should appear at Centro rather than at Belisario. Evidence on the full spatial extent of the effect remains an open question.

5 Results

The results support one main claim: Metro Line 1 is associated with a localized reduction in fine-particulate pollution at Quito’s historic center, where the urban form makes car access difficult and metro access especially valuable. The evidence is strongest for $PM_{2.5}$ at Centro. Belisario, which is almost as close to the metro corridor but outside the historic center, does not show the same pattern. This contrast points to a localized particulate effect at the predicted destination, rather than a generic proximity effect along the corridor. The results do not establish the full spatial extent of the effect, and they do not by themselves prove the traffic channel.

The empirical prediction is specific. Evidence for a destination-access mechanism should show a $PM_{2.5}$ decline at Centro, no comparable decline at Belisario, and no similar pattern across the rest of the monitoring network. Evidence for a simple corridor-proximity mechanism would instead predict similar declines at both Centro and Belisario. We organize the results around this distinction.

5.1 Local $PM_{2.5}$ effects at Centro

Figure 5 shows the weekly event-study estimates for the preferred Centro specification. The pre-opening gaps fluctuate around zero, while the post-opening estimates move below zero during the first months of operation and deepen through 2024. During the early post-opening period before the blackout and wildfire disruptions, the average effect is -0.130 log points. The estimate is nearly identical, -0.132 log points, when later post-opening weeks are added but the disruption weeks are excluded. The figure therefore shows that the preferred estimate is not driven by a single

post-opening week or by the late-2024 disruption. The gradual deepening of the effect is consistent with a response that builds as metro ridership grows rather than a discrete shift at opening.

Table 5 reports the local $PM_{2.5}$ estimates. The table is organized around the identification logic. The first column treats Centro and Belisario jointly as the two monitoring stations closest to Line 1. This proximity-based specification produces negative estimates in the clean windows, but the effects are not statistically significant. This suggests that the $PM_{2.5}$ result is concentrated at the predicted historic-center destination rather than simply tracing proximity to the metro corridor.

The second column treats Belisario as the affected station and excludes Centro from the donor pool. Belisario is almost as close to the metro corridor as Centro, but its estimates are small, imprecise, and not statistically significant in the clean windows. This first establishes Belisario as a placebo-treated station with no comparable response, which in turn supports using it as a valid donor for Centro.

The third column treats Centro as the affected station and removes Belisario from the donor pool. This conservative specification produces larger negative point estimates, 17.0 percent in the pre-disruption window and 14.2 percent in the donut window, but the estimates are imprecise. The lack of precision is informative: removing Belisario also removes the closest and most useful comparison station for Centro.

The preferred specification, in the final column, treats Centro as the affected station and retains Belisario in the donor pool. This specification estimates a 12.2 percent reduction in $PM_{2.5}$ in the pre-disruption window, with a two-sided conformal p -value of 0.011. The estimate is nearly unchanged in the donut window, at 12.4 percent. The full post-opening window gives a larger reduction of 15.1 percent, but because it includes the late-2024 disruption, we interpret the pre-disruption estimate as the cleanest local effect.

The preferred specification delivers the statistically significant Centro effect while also providing the strongest pre-opening fit. Its pre-treatment RMSPE is 0.129, compared with 0.185 to 0.205 in the other specifications, and its post-to-pre RMSPE ratio rises to 1.98 in the pre-disruption window. This means that retaining Belisario improves the counterfactual fit before treatment and reveals a post-opening divergence that is large relative to the pre-opening prediction error. The next section broadens the test to the full monitoring network, asking whether the Centro result is unique among Quito's monitors and whether other stations show comparable evidence of a post-opening $PM_{2.5}$ reduction.

5.2 Spatial placebo evidence

The spatial placebo exercise broadens the Centro-Belisario comparison to the full monitoring network. We treat each REMMAQ station in turn as if it were affected, construct a synthetic counterfactual from the remaining stations, and apply the same two-sided conformal inference procedure. This

Table 5. Local PM_{2.5} identification sequence

	Centro + Belisario co-treated	Belisario treated Centro excluded	Centro treated Belisario excluded	Centro treated Belisario retained
<i>Panel A. Pre-disruption window: Nov. 27, 2023 to Sept. 9, 2024</i>				
Log effect	-0.124	-0.062	-0.186	-0.130
Percent effect	-11.7	-6.0	-17.0	-12.2
Conformal <i>p</i>	0.610	0.729	0.392	0.011
Pre RMSPE	0.185	0.188	0.205	0.129
Post RMSPE	0.230	0.218	0.308	0.255
Post/pre RMSPE	1.24	1.16	1.50	1.98
<i>Panel B. Donut window: excluding Sept. 16 to Dec. 16, 2024</i>				
Log effect	-0.089	-0.024	-0.153	-0.132
Percent effect	-8.5	-2.3	-14.2	-12.4
Conformal <i>p</i>	0.809	0.849	0.486	0.041
Pre RMSPE	0.185	0.188	0.205	0.129
Post RMSPE	0.253	0.259	0.308	0.244
Post/pre RMSPE	1.37	1.38	1.50	1.89
<i>Panel C. Full post-opening window: Nov. 27, 2023–Mar. 2025</i>				
Log effect	0.040	0.140	-0.060	-0.164
Percent effect	4.1	15.1	-5.8	-15.1
Conformal <i>p</i>	0.807	0.698	0.506	0.006
Pre RMSPE	0.185	0.188	0.205	0.129
Post RMSPE	0.581	0.671	0.534	0.272
Post/pre RMSPE	3.14	3.58	2.61	2.10

Notes. All specifications use augmented synthetic control with station fixed effects and meteorological controls. Effects are reported in logs and percent terms. *p*-values are two-sided conformal *p*-values. The preferred specification is the final column: Centro treated, Belisario retained in the donor pool. The table is ordered to distinguish a corridor-proximity interpretation from a destination-access interpretation. Belisario is first tested as a treated station with Centro excluded; after showing no comparable effect at Belisario, the preferred specification retains Belisario as the key donor for Centro.

exercise is not used as the main source of statistical significance. With only eight stations, any rank-based statistic is mechanically coarse. Instead, the exercise is a spatial diagnostic: it asks whether the significant PM_{2.5} effect appears at the station predicted by the design, and whether other stations show comparable evidence of a post-opening reduction.

Table 6 reports the station-level placebo estimates for the pre-disruption window. Because the placebo re-estimates every station, including Centro, through the same leave-one-out routine, the Centro estimate differs trivially from the configured estimate in Table 5. Centro is the only station with a statistically significant negative effect. Its estimated reduction is 12.2 percent, with a two-sided conformal *p*-value of 0.012. Belisario, despite being almost equidistant from the metro corridor, has a positive and statistically insignificant estimate of 8.5 percent. This connects directly to the identification sequence in Table 5: the same station that anchors Centro's preferred counterfactual does not show a comparable response when treated as the exposed station.

The remaining stations do not show comparable evidence of a post-opening PM_{2.5} reduction. The only other estimate approaching conventional significance is Cotocollao, but its point estimate is positive, so it is not a competing pollution-reduction result. Tumbaco has a large negative point

Table 6. Spatial placebo estimates for PM_{2.5}

Station	Distance (km)	Effect (%)	Conformal p -value
Centro	0.61	-12.2	0.012
Belisario	0.77	8.5	0.452
Guamaní	4.04	-7.8	0.523
Cotocollao	5.21	4.5	0.061
Carapungo	7.64	-8.9	1.000
Los Chillos	8.62	21.8	0.135
Tumbaco	9.56	-17.7	0.785
San Antonio	16.65	9.2	0.574

Notes. Entries report leave-one-station-out placebo estimates for weekly peak-hour PM_{2.5} in the pre-disruption window. Each row treats the listed station as if it were exposed and uses the remaining stations as the donor pool. Effects are estimated using augmented synthetic control with station fixed effects and meteorological controls. Distance is the straight-line distance from the monitoring station to the nearest Metro Line 1 stop. The p -values are two-sided conformal p -values. The rank-based placebo statistic is not used as the inferential test because the local placebo sample contains only eight stations. We also do not rank stations by post-to-pre fit ratios, because with eight stations those ratios can be sensitive to a single station's pre-period noise; the conformal procedure instead uses the full sequence of prediction errors.

estimate, but its two-sided conformal p -value is 0.785, meaning that its post-opening deviation is not statistically distinguishable from its pre-opening prediction errors. The evidence therefore supports a localized effect at the historic center rather than a simple distance-to-corridor interpretation.

This result should not be read as evidence that the metro had no effects elsewhere. The monitoring network has only eight stations, the post-opening period is short, ridership was still growing, and the late-2024 disruption reduces power to detect broader spatial responses. The local design is strongest for the destination-access test: whether the particulate response appears at Centro and not at the almost equidistant non-historic center station.

5.3 Local pollutant pattern

Table 7 reports the preferred local specification for all four ground-monitored pollutants. The pattern is particulate-specific. PM_{2.5} falls significantly in every window, with the cleanest estimate in the pre-disruption period. NO₂ provides the cleanest null result. It has the best pre-treatment fit of any pollutant in the table, with a pre-RMSPE of 0.089, and its post-to-pre RMSPE ratio is 1.02 in the pre-disruption window. The synthetic counterfactual therefore tracks NO₂ tightly before treatment and shows essentially no post-opening divergence. CO is different: it also has negative point estimates, but the conformal p -values equal 1.000 and the post-to-pre RMSPE ratios are below one, so the CO estimates do not provide evidence of a post-opening divergence.

SO₂ behaves differently. It is close to zero and statistically insignificant in the pre-disruption window, but turns positive once later post-opening weeks are included. The donut estimate is 13.1 percent, with a two-sided conformal p -value of 0.055, and the full-window estimate is 10.0 percent. This pattern is consistent with the diesel-combustion signature expected from widespread generator use during the late-2024 energy crisis. Rather than supporting a metro effect, the SO₂

Table 7. Local pollutant effects at Centro

	PM _{2.5}	CO	NO ₂	SO ₂
<i>Panel A. Pre-disruption window</i>				
Log effect	-0.130	-0.056	-0.064	-0.025
Percent effect	-12.2	-5.4	-6.2	-2.5
Conformal p	0.011	1.000	0.967	0.187
Pre RMSPE	0.129	0.248	0.089	0.214
Post RMSPE	0.255	0.163	0.090	0.293
Post/pre RMSPE	1.98	0.66	1.02	1.37
<i>Panel B. Donut window</i>				
Log effect	-0.132	-0.082	-0.055	0.123
Percent effect	-12.4	-7.9	-5.4	13.1
Conformal p	0.041	1.000	0.610	0.055
Pre RMSPE	0.129	0.248	0.089	0.214
Post RMSPE	0.244	0.168	0.138	0.403
Post/pre RMSPE	1.89	0.68	1.56	1.89
<i>Panel C. Full post-opening window</i>				
Log effect	-0.164	-0.108	-0.060	0.095
Percent effect	-15.1	-10.2	-5.8	10.0
Conformal p	0.006	1.000	0.716	0.124
Pre RMSPE	0.129	0.248	0.089	0.214
Post RMSPE	0.272	0.189	0.131	0.381
Post/pre RMSPE	2.10	0.76	1.48	1.78

Notes. All columns use the preferred local specification: Centro treated, Belisario retained in the donor pool, augmented synthetic control with station fixed effects and meteorological controls. Effects are reported in logs and percent terms. p -values are two-sided conformal p -values. Panel A is the primary local window and covers the post-opening period before the late-2024 blackout and wildfire disruption, from Nov. 27, 2023 to Sept. 9, 2024. Panel B excludes the disruption weeks, Sept. 16 to Dec. 16, 2024. Panel C retains all available post-opening weeks, through Mar. 2025 for PM_{2.5} and through June 2025 for CO, NO₂, and SO₂. Pre RMSPE is identical across panels within each pollutant because all three windows use the same pre-treatment period. PM_{2.5} is available for eight stations. CO, NO₂, and SO₂ are available for seven stations because San Antonio does not report these pollutants.

results corroborate the decision to isolate the pre-disruption window as the primary local estimand.

Taken together, the local pollutant results point to a particulate-specific response rather than a broad reduction across all pollutants. This pattern is consistent with changes in local traffic activity, congestion, resuspended road dust, or other activity around the historic center, but it does not identify the exact emissions channel. The absence of comparable reductions in CO and NO₂ means that the direct traffic mechanism should be tested with ridership, congestion, vehicle-flow, or parking data rather than inferred from the pollutant pattern alone.

The PM_{2.5} result is qualitatively robust to alternative estimators. Augmented synthetic control without station fixed effects gives reductions of 9.5 percent in the pre-disruption window, 10.1 percent in the donut window, and 13.3 percent in the full window, all significant under two-sided conformal inference. Synthetic difference-in-differences gives larger negative point estimates, ranging from

14.3 to 20.7 percent across windows, with a pre-disruption estimate of 14.5 percent. Because valid local inference was not computed for SDID, we use these estimates only as robustness checks for sign and magnitude. We report the full specification set in the appendix [A.2](#).

5.4 Citywide satellite estimates

The satellite analysis is not designed to identify the historic-center mechanism directly. It asks a broader question: whether Quito as an urban area experienced a detectable aggregate change in pollution relative to comparable Latin American urban areas. Table 8 reports the citywide SDID estimates for AOD, the satellite particulate proxy, across donor pools. The estimates are negative across all donor pools and windows. In the preferred 100-city donor pool, selected using the pre-specified Mahalanobis-distance rule, AOD falls by 0.113 log points during the first months of operation before the blackout and wildfire disruptions. The corresponding two-sided Wald p -value from placebo standard errors is 0.035.

The pool-walk supports the same conclusion. In the primary pre-disruption window, the AOD estimate is negative in every donor pool, ranging from -0.100 to -0.145 log points. The 95 percent confidence intervals exclude zero in every pool, although the upper bound in the 50-city pool sits essentially at zero, indicating that precision is weakest in the most restrictive donor pool. The donut and full-window estimates are also negative and generally larger in magnitude. The result therefore does not depend on selecting the 100-city pool because it is significant; that pool is the preferred specification because it follows the pre-specified donor-selection rule.

The AOD results point in the same particulate direction as the local $PM_{2.5}$ evidence, but they should not be interpreted as a clean citywide metro effect. The satellite estimand is different from the local station estimand. AOD measures aerosol loading in the atmospheric column rather than surface $PM_{2.5}$, and the broader Ecuadorian environment in 2024 may have affected Quito differently from other Latin American cities. The magnitude of the citywide AOD estimates is also large relative to what a single metro line would likely produce at the urban scale. For these reasons, we interpret the satellite analysis as a bounded regional benchmark: it corroborates the direction of the particulate result, but it is not the main causal test.

Table 9 reports the preferred 100-city SDID specification for all satellite outcomes. The particulate proxy is the only outcome that moves consistently. CO is small and statistically insignificant in all windows. NO₂ changes sign across windows and is not statistically significant. SO₂ is negligible in magnitude. These gas-pollutant estimates are shown for transparency, but they are not load-bearing for the mechanism. At the urban-center scale, weekly TROPOMI gas retrievals average over the entire city polygon and are affected by atmospheric transport, retrieval noise, vertical-column measurement, and non-transport sources. Their null or inconsistent estimates therefore limit what the satellite analysis can say about channels, but they do not overturn the local $PM_{2.5}$ evidence.

Table 8. Citywide AOD estimates across donor pools

Donor pool	Pre-disruption	Donut	Full
50 nearest cities	-0.100 (0.049) [-0.200, -0.000]	-0.089 (0.247) [-0.240, 0.062]	-0.107 (0.142) [-0.250, 0.036]
100 nearest cities [†]	-0.113 (0.035) [-0.218, -0.008]	-0.171 (0.016) [-0.309, -0.032]	-0.188 (0.006) [-0.323, -0.054]
150 nearest cities	-0.145 (0.008) [-0.252, -0.038]	-0.133 (0.099) [-0.290, 0.025]	-0.155 (0.037) [-0.301, -0.010]
200 nearest cities	-0.138 (0.016) [-0.250, -0.026]	-0.156 (0.018) [-0.286, -0.027]	-0.175 (0.035) [-0.338, -0.013]
All balanced cities	-0.143 (0.011) [-0.253, -0.033]	-0.170 (0.009) [-0.299, -0.042]	-0.185 (0.006) [-0.318, -0.053]

Notes. Entries are log-point effects on AOD from synthetic difference-in-differences. Two-sided p -values from placebo standard errors using the normal approximation are in parentheses; 95 percent confidence intervals are in brackets. Donor pools are ranked by Mahalanobis distance to Quito using pre-treatment and time-invariant city characteristics. The pre-disruption window is the primary citywide window because it uses the post-opening period before the late-2024 blackout and wildfire disruption. [†]The preferred citywide specification uses the 100 nearest donor cities, following the pre-specified Mahalanobis-distance donor-selection rule. Permutation p -values are retained as a secondary appendix check and lead to the same substantive interpretation, with weaker precision in the smallest donor pools.

The results section therefore indicates a localized particulate response at Quito’s historic center. PM_{2.5} falls at Centro, the station predicted by the destination-access mechanism, while Belisario, the similarly close non-center station, does not show the same pattern. The satellite AOD estimates move in the same particulate direction at the city scale, but the satellite gases are not informative enough to establish a mechanism. We therefore interpret the findings as evidence that Quito Metro Line 1 reduced fine-particulate pollution at the historic center, consistent with destination-based mode substitution, while leaving the full spatial extent and direct traffic channel for future mobility evidence.

6 Conclusion

This paper studies whether the air-quality effects of a new metro line can be concentrated within a city. Using the opening of Quito Metro Line 1, we compare pollution changes at the historic center with changes at other monitoring stations and then place the local evidence in a broader citywide comparison using satellite data. The main result is a localized decline in fine particulate pollution at Centro, the monitoring station inside Quito’s historic center. PM_{2.5} falls significantly, by about 12 percent, during the first months of operation before Ecuador’s late-2024 blackout and wildfire disruptions. The effect is not mirrored at Belisario, a station that is nearly as close to the metro corridor but outside the historic center.

The spatial pattern is central to the interpretation. The results do not support a simple proximity story in which pollution falls mechanically near the rail corridor. Instead, the evidence points to a destination-based response: the metro appears to matter most where it changes access to a dense

Table 9. Citywide satellite estimates by pollutant

	AOD	CO	NO ₂	SO ₂
<i>Panel A. Pre-disruption window</i>				
Effect	-0.113	-0.018	0.034	-0.000
Wald <i>p</i>	0.035	0.303	0.571	0.250
<i>Panel B. Donut window</i>				
Effect	-0.171	-0.013	-0.033	-0.000
Wald <i>p</i>	0.016	0.310	0.606	0.292
<i>Panel C. Full post-opening window</i>				
Effect	-0.188	-0.005	-0.032	-0.000
Wald <i>p</i>	0.006	0.710	0.606	0.079

Notes. Entries are treatment-effect estimates from the preferred 100-city SDID specification. *p*-values are two-sided *p*-values from placebo standard errors using the normal approximation. AOD, CO, and NO₂ are log-transformed, so coefficients are log-point effects. SO₂ is transformed using the inverse hyperbolic sine; its estimated magnitudes are close to zero. The pre-disruption window is the primary citywide window. Permutation *p*-values are retained as a secondary appendix check and lead to the same substantive interpretation.

area in which car use is already costly. This distinction matters because urban rail investments are often evaluated as citywide infrastructure projects, while their environmental benefits may be highly local. A modest or ambiguous citywide effect can coexist with meaningful gains in places where the transport intervention changes the relative attractiveness of driving.

The satellite analysis supports this interpretation only in a limited sense. Aerosol optical depth moves in the same particulate direction as the local PM_{2.5} result, while satellite gas outcomes do not show a consistent pattern. We therefore interpret the citywide evidence as complementary, not as the paper’s main causal test. Satellite measures capture pollution at a different spatial and atmospheric scale, and Quito’s broader post-opening period coincided with national shocks that complicate citywide comparisons. The strongest evidence remains the station-level result at the location where the mechanism predicts the clearest early effect.

These findings have two implications for policy. First, large transit investments may not generate broad environmental benefits on their own. Their local air-quality effects are more likely to appear where the new system makes car use less attractive relative to public transport. In dense destinations, especially places with limited parking, constrained street networks, and high pedestrian activity, rail access may produce cleaner local environments even before broader citywide changes are visible. Second, transport policy may need to pair supply-side investments, such as metros, with complementary measures that limit the convenience of driving in the areas where modal shift is most valuable.

The paper also points to the limits of the current evidence. The results document where the early particulate response appears, but they do not directly measure the traffic channel. Future work

should combine air-quality data with ridership, congestion, vehicle counts, parking behavior, and origin-destination travel patterns to test whether the observed pollution changes reflect reduced vehicle access to the historic center, changes in bus circulation, lower congestion, or other local activity responses. As Quito's metro matures and ridership grows, these data will be important for understanding whether the early localized gains expand, persist, or depend on additional policies that reshape travel behavior.

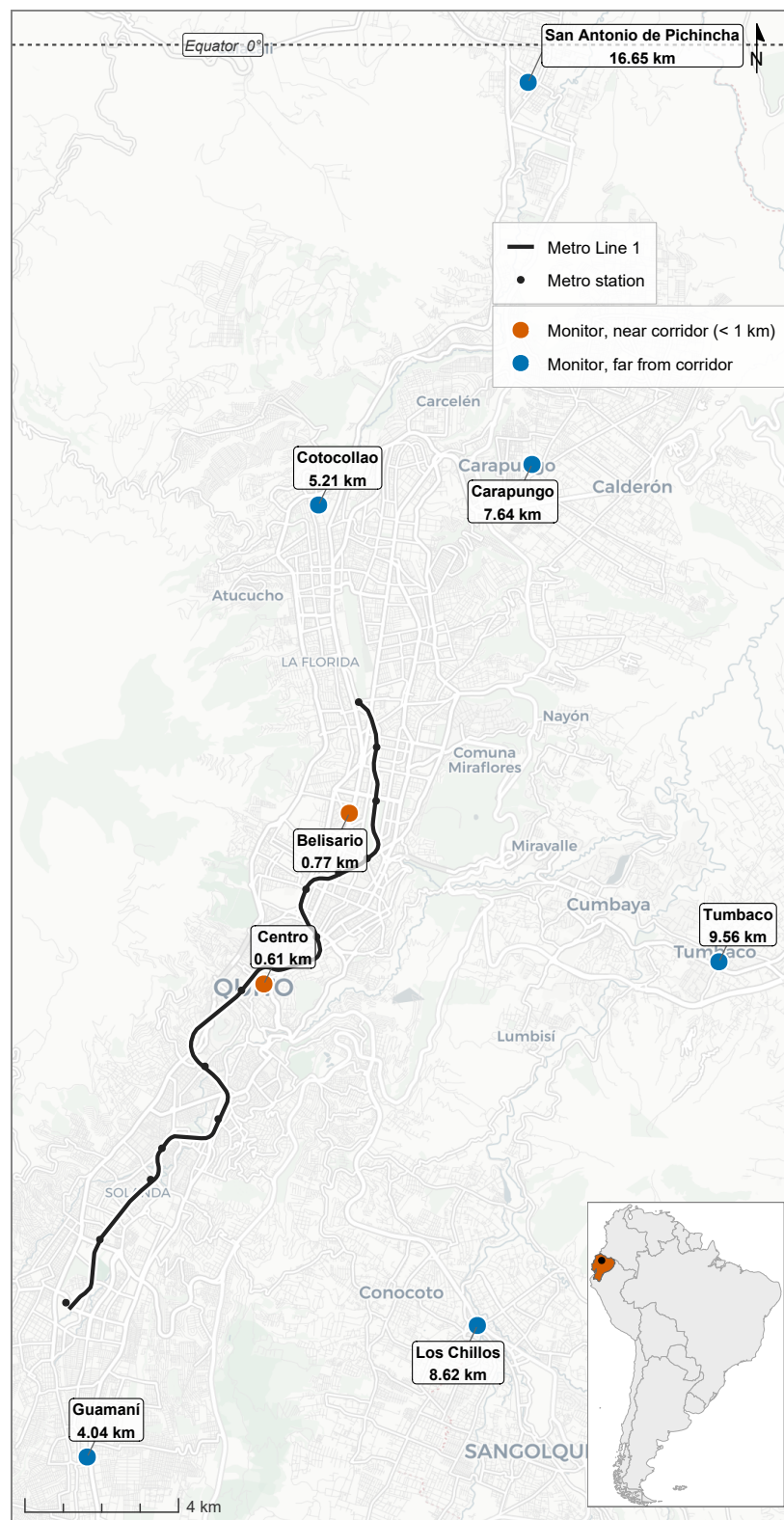


Figure 1. Air-quality monitoring stations and Metro Line 1. The black line traces Metro Line 1 along Quito's north-south valley corridor; black dots mark its stations. Orange dots mark the two REMMAQ monitors within 1 km of the corridor, Centro and Belisario; blue dots mark the six monitors farther from the corridor. Labels give each monitor's straight-line distance to the nearest metro stop. The inset locates Quito within South America.

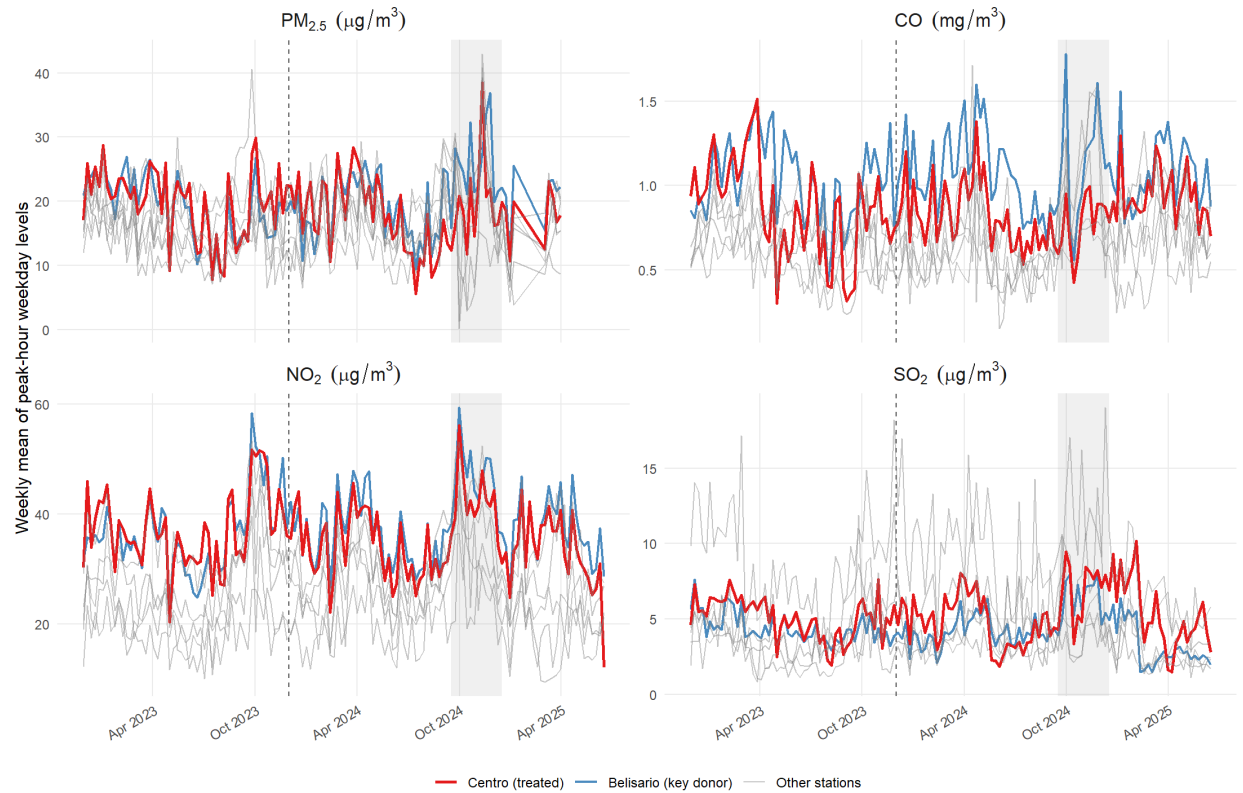


Figure 2. Weekly peak-hour pollution at REMMAQ monitoring stations. The figure plots weekday peak-hour weekly means for $PM_{2.5}$, CO, NO_2 , and SO_2 . Centro, the treated station, is shown in red; Belisario is shown in blue; all other stations are shown in gray. The vertical dashed line marks the opening of Metro Line 1 on December 1, 2023. The shaded region marks the Phase 3 power-outage and wildfire disruption weeks used to define the donut window.

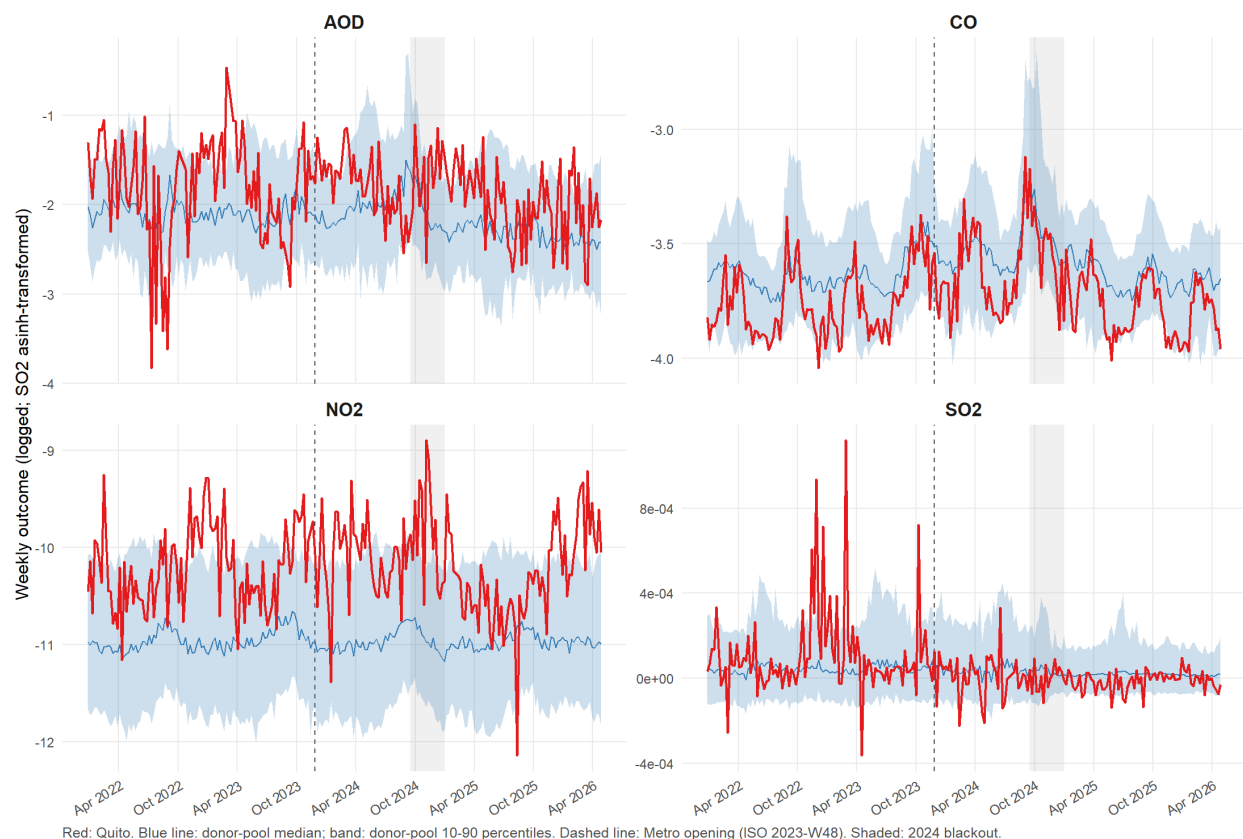


Figure 3. Weekly satellite outcomes for Quito and the donor pool. Red lines plot Quito. Blue lines plot the donor-pool median, and shaded bands show the 10th to 90th percentiles of the donor pool. The vertical dashed line marks the Metro Line 1 opening week, ISO 2023-W48. The shaded region marks the 2024 disruption period.



Figure 4. Satellite coverage over time. Blue lines show the percentage of donor-pool cities with a valid observation in each week. Red ticks mark weeks with no valid raw observation for Quito before short-gap imputation. The vertical dashed line marks the Metro Line 1 opening week, ISO 2023-W48.

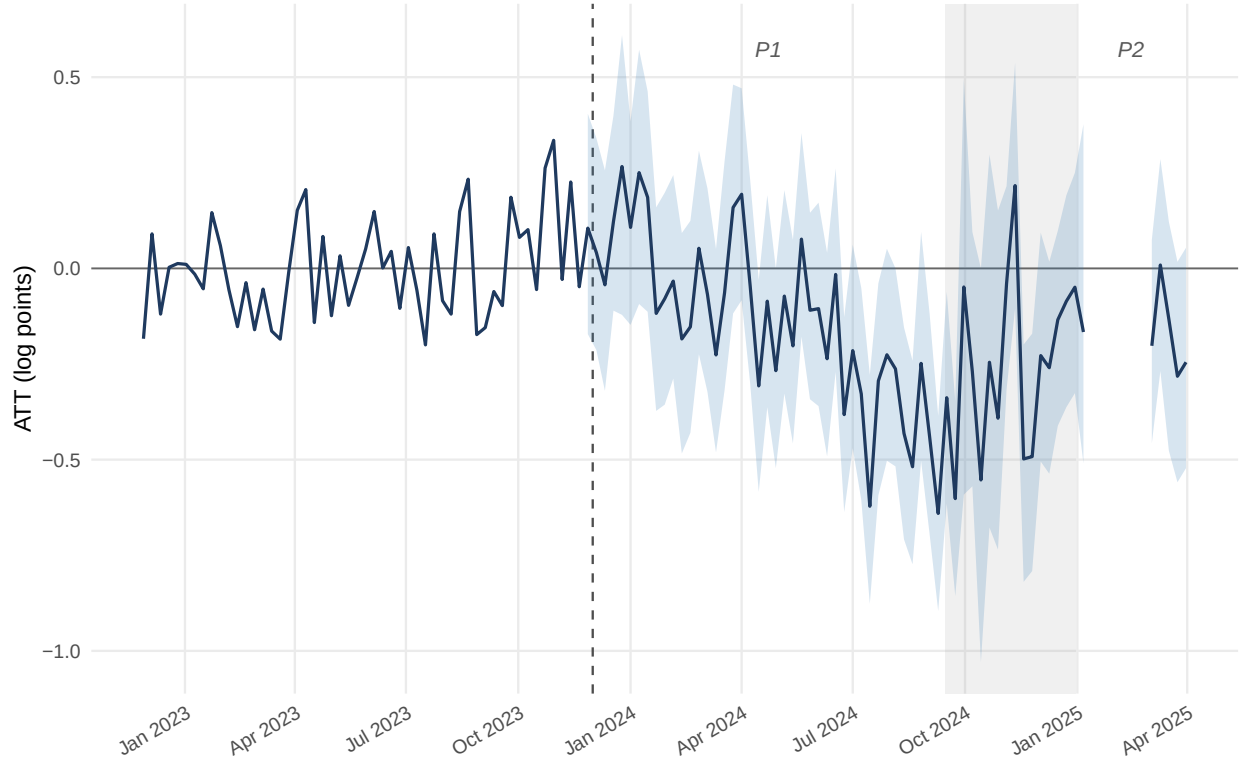


Figure 5. Effect of Quito Metro Line 1 on $PM_{2.5}$: event-study estimates.

Notes. The figure reports estimates from augmented synthetic control with station fixed effects and weather covariates. The solid line shows the weekly treatment effect in log points, and the shaded ribbon shows the pointwise conformal confidence interval. The vertical dashed line marks the opening of Quito Metro Line 1 in December 2023. The gray region marks the blackout and wildfire disruption period. Average treatment effects by window are: pre-disruption, P_1 , -0.130 log points ($p = 0.011$); donut, $P_1 + P_2$, -0.132 log points ($p = 0.041$); and full post-opening, -0.164 log points ($p = 0.006$). All p -values are two-sided conformal p -values. Period 2 (post-disruption) contains only 6 $PM_{2.5}$ weeks: data are absent for all stations from the week of 13 Jan to the week of 24 Feb 2025, so the panel jumps from the week of 6 Jan 2025 to the week of 3 Mar 2025. The break in the series in early 2025 marks this gap.

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Table A.1. Local sample construction

Pollutant	Stations	Raw hourly	Peak-hour	Weekday peak-hour	Balanced panel	Imputed (%)
PM _{2.5}	8	198,778	49,697	35,427	928	3.0
CO	7	167,214	43,457	30,973	938	1.0
NO ₂	7	176,580	45,622	32,560	938	0.7
SO ₂	7	177,295	45,378	32,370	938	1.3

The table reports non-missing observations at each step. Peak hours are 7–9 a.m. and 5–7 p.m.; weekday peak-hour observations restrict these to Monday through Friday. Balanced panel counts are station-week observations after weekly aggregation and limited imputation. The PM_{2.5} panel spans 116 weeks; the gas panels span 134 weeks.

A Additional Data Construction Details

A.1 Satellite coverage, imputation, and balance diagnostics

Table A.2. Satellite pre-treatment diagnostics for the preferred 100-city donor pool

Variable	AOD	CO	NO ₂	SO ₂
Transformed outcome	0.196	-0.386	0.616	-0.050
Calm hours	1.688	1.294	1.573	1.684
Evaporation	0.224	0.217	0.259	0.249
Rain frequency	2.242	1.739	1.971	2.025
Relative humidity	1.413	1.219	1.353	1.384
Soil temperature	-2.364	-1.935	-2.405	-2.540
Soil moisture	1.714	1.524	1.657	1.685
Solar radiation	-1.365	-1.269	-1.333	-1.352
Surface pressure	-2.433	-2.036	-2.568	-2.642
Temperature	-2.465	-2.077	-2.569	-2.652
Precipitation	1.656	1.277	1.353	1.369
Wind direction	2.140	2.162	2.152	2.158
Maximum wind speed	-1.647	-1.290	-1.502	-1.638
Mean wind speed	-1.071	-1.195	-1.169	-1.217

Notes. Entries are normalized pre-treatment differences between Quito and the preferred 100-city donor pool. Outcomes are transformed as in the estimation: log for AOD, CO, and NO₂, and inverse hyperbolic sine for SO₂. Weather covariates are weekly ERA5-Land controls. These diagnostics describe the raw donor pool before estimator weighting and residualization, so they should be interpreted as sample diagnostics rather than as final synthetic-control fit.

A.2 Full local PM_{2.5} specification battery

This appendix reports the complete local PM_{2.5} specification battery behind the identification sequence in Table 5 and the robustness statement in the main text. The battery crosses three estimators, synthetic difference-in-differences and augmented synthetic control with and without station fixed effects, with four treated-unit configurations and three post-opening windows. The main pattern is clear: the Centro-treated, Belisario-retained configuration is the only one that combines a tight pre-treatment fit with statistically significant negative estimates across the augmented synthetic control specifications. The synthetic difference-in-differences estimates are also negative for this

configuration, but we do not report inferential tests for the local SDID estimates because neither large-sample approximations nor donor-placebo permutation tests are credible with this small monitoring network, and conformal inference was not computed for SDID.

Table A.3. Local PM_{2.5}: full specification battery across estimators, treated-unit configurations, and windows

M	Configuration	Log effect, p -value beneath			Pre-disruption fit	
		Pre-disruption [†]	Donut	Full	RMSPE _{pre}	Post/pre
<i>Panel A. Synthetic difference-in-differences</i>						
M1	Centro + Belisario co-treated	-0.123 n.a.	-0.092 n.a.	0.005 n.a.	0.261	0.95
M3	Belisario treated, Centro dropped	-0.038 n.a.	0.001 n.a.	0.118 n.a.	0.258	0.88
M2	Centro treated, Belisario dropped	-0.208 n.a.	-0.184 n.a.	-0.108 n.a.	0.281	1.13
M2b	Centro treated, Belisario retained	-0.157 n.a.	-0.154 n.a.	-0.232 n.a.	0.234	1.23
<i>Panel B. Augmented synthetic control, no station fixed effects</i>						
M4	Centro + Belisario co-treated	-0.063 (0.906)	-0.033 (0.952)	0.084 (0.860)	0.198	0.98
M6	Belisario treated, Centro dropped	-0.006 (0.871)	0.028 (0.884)	0.177 (0.670)	0.195	0.95
M5	Centro treated, Belisario dropped	-0.121 (0.836)	-0.095 (0.858)	-0.007 (0.764)	0.221	1.24
M5b	Centro treated, Belisario retained	-0.100 (0.011)	-0.106 (0.020)	-0.143 (0.003)	0.132	1.83
<i>Panel C. Augmented synthetic control with station fixed effects</i>						
M7	Centro + Belisario co-treated	-0.124 (0.610)	-0.089 (0.809)	0.040 (0.807)	0.185	1.24
M9	Belisario treated, Centro dropped	-0.062 (0.729)	-0.024 (0.849)	0.140 (0.698)	0.188	1.16
M8	Centro treated, Belisario dropped	-0.186 (0.392)	-0.154 (0.486)	-0.060 (0.506)	0.205	1.50
M8b	Centro treated, Belisario retained[‡]	-0.130 (0.011)	-0.132 (0.041)	-0.164 (0.006)	0.129	1.98

Notes. Each cell reports the treatment effect on log weekly peak-hour weekday PM_{2.5}, with the two-sided *p*-value in parentheses beneath when valid inference is available. Every specification is re-estimated separately on each window. Rows are labeled by treated-unit configuration; the *M* code is retained for traceability and matches the estimation script.

[†] Pre-disruption is the primary window, ending before the onset of the 2024 electricity-rationing and wildfire disruption. RMSPE_{pre} is the pre-treatment root mean squared prediction error. The post-to-pre RMSPE ratio is reported for the pre-disruption window only. A ratio near or below one indicates little post-period divergence from the synthetic control, which is why the co-treated and Belisario-treated specifications are insignificant despite, in some cases, sizable point estimates.

[‡] Preferred specification: augmented synthetic control with station fixed effects, Centro treated with Belisario retained in the donor pool. This row is also the final column of the main identification table, Table 5.

Panel A reports SDID point estimates only. We do not report *p*-values for the local SDID estimates because large-sample approximations are not credible with this small monitoring network, donor-placebo permutation inference is mechanically coarse, and conformal inference was not computed for SDID. The SDID estimates are therefore used only to assess robustness of sign and magnitude.

Panels B and C use two-sided conformal inference. Co-treated specifications pool Centro and Belisario into a single treated aggregate. Belisario-treated specifications place treatment on the more distant Belisario station and serve as placebo-style checks.

All specifications condition on meteorology. The augmented synthetic control specifications adjust for meteorological covariates within the estimator. Synthetic difference-in-differences uses the meteorology-residualized log outcome. The fixed-effects distinction applies only to the augmented synthetic control panels.